



Qualification Pack



Tunnel Safety Supervisor

QP Code: ICE/CON/Q0601

Version: 1.0

NSQF Level: 5

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ICE/CON/Q0601: Tunnel Safety Supervisor

Brief Job Description

The Tunnel Safety Supervisor is responsible for ensuring the safety and regulatory compliance of tunnel operations. This role includes conducting regular safety inspections and audits to identify and mitigate potential hazards, ensuring adherence to safety standards and regulations. The supervisor develops and maintains emergency response plans, conducts safety drills and provides training to personnel on emergency procedures. Additionally, they coordinate with various teams to implement safety measures, continuously assess risks and take corrective actions to prevent accidents and ensure a safe working environment.

Personal Attributes

Teamwork

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ICE/CON/N0601: Implement Tunnel Site Safety Management System](#)
2. [ICE/CON/N0602: Conduct Hazard Identification, Risk Assessment & Control for Tunnelling](#)
3. [ICE/CON/N0603: Manage Ventilation, Gas Monitoring and Occupational Hygiene in Tunnels](#)
4. [ICE/CON/N0604: Emergency Preparedness, Rescue & Evacuation in Tunnel Works](#)
5. [ICE/CON/N0605: Manage Ground Support, Excavation Face Safety and Rockfall Controls](#)
6. [ICE/CON/N0606: Manage Plant, TBM, Mechanical & Blasting Safety in Tunnelling](#)
7. [ICE/CON/N9901: Implement Safe Work Practices and Environmental Stewardship at Construction Sites](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	Real Estate and Infrastructure Construction
Occupation	Safety works



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Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3257.0600
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (B.E. / B.Tech in Civil Engineering) with 4 Years of experience Relevant Industry OR 12th grade Pass with 5 Years of experience Relevant Industry OR 10th grade pass with 6 Years of experience Relevant Industry OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Relevant Industry
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Not Applicable
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	NG-05-CO-046462025-V1-ICES
NQR Version	1.0

Remarks:

Min. Job Entry Age will be as per Govt. Norms



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ICE/CON/N0601: Implement Tunnel Site Safety Management System

Description

This NOS describes how to implement and monitor the site safety management system for tunnelling works, ensuring compliance with project safety plans, regulatory requirements and company safety procedures.

Scope

The scope covers the following :

- Establish safety management arrangements for tunnel works
- Monitor compliance with safety procedures and statutory requirements
- Safety documentation and record-keeping
- Training, competency and toolbox talks
- Contractor/subcontractor safety management

Elements and Performance Criteria

Establish Safety Management Arrangements for Tunnel Works

To be competent, the user/individual on the job must be able to:

- PC1.** confirm and implement the project Tunnel Safety Management Plan (TSMP) as issued by project management
- PC2.** ensure roles, responsibilities and communication lines for safety on the tunnelling site are documented and displayed
- PC3.** coordinate with project EHS and engineering teams to integrate TSMP with construction programme
- PC4.** verify that required permits, licenses and statutory registers are available on site
- PC5.** ensure safety briefings and induction schedules are planned and delivered for all incoming personnel

Monitor Compliance with Safety Procedures and Statutory Requirements

To be competent, the user/individual on the job must be able to:

- PC6.** conduct routine site safety inspections and record non-conformances
- PC7.** ensure implementation of corrective and preventive actions within agreed timeframes
- PC8.** verify the use and condition of safety equipment and PPE as per TSMP
- PC9.** report incidents, near misses and unsafe acts/conditions as per reporting procedures
- PC10.** liaise with regulators and project QA/EHS for audits and inspections

Safety Documentation and Record-Keeping

To be competent, the user/individual on the job must be able to:

- PC11.** maintain daily safety logs, gas monitoring records, ventilation logs and permits (confined space, hot work, LOTO, blasting)
- PC12.** ensure accurate attendance, competency and medical fitness records for tunnel workforce
- PC13.** manage issuance, review and archival of method statements and risk assessments



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- PC14.** prepare and submit periodic safety performance reports to site management.
- PC15.** ensure signage, hazard maps and emergency contact lists are current and accessible

Training, Competency and Toolbox Talks

To be competent, the user/individual on the job must be able to:

- PC16.** identify safety training needs and ensure competence verification before assignment
- PC17.** schedule and conduct site-specific toolbox talks and safety drill
- PC18.** ensure specialist training (confined space, breathing apparatus, gas monitoring, emergency rescue) is current
- PC19.** monitor on-the-job coaching and corrective instruction for unsafe behaviour
- PC20.** maintain training matrix and certificates for audit

Contractor/Subcontractor Safety Management

To be competent, the user/individual on the job must be able to:

- PC21.** verify that subcontractors submit safety method statements and risk assessments before commencement
- PC22.** conduct pre-start safety meetings with subcontractor supervisors
- PC23.** monitor subcontractor adherence to TSMP and take action on non-compliance
- PC24.** ensure coordination of interfaces (traffic, hoisting, muck handling) to prevent cross-contractor hazards
- PC25.** enforce contractor performance metrics and include safety KPIs in reviews

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legal and regulatory framework for construction and tunnelling safety (national statutes and local permits)
- KU2.** key contents and objectives of a Tunnel Safety Management Plan (TSMP)
- KU3.** duties and responsibilities of employer, contractor, supervisor and worker under applicable laws
- KU4.** permit systems: confined space, hot work, LOTO, entry/exit controls, blasting permits
- KU5.** incident reporting, investigation, root cause analysis and corrective action processes
- KU6.** record keeping and documentation requirements for gas monitoring, ventilation, training and permits
- KU7.** principles of behavioural safety and toolbox talk delivery
- KU8.** requirements for medical fitness, PPE selection and respiratory protective equipment
- KU9.** contractor management and interface coordination principles
- KU10.** safety auditing and inspection techniques
- KU11.** basic risk management and hazard identification methods
- KU12.** emergency preparedness basics relevant to the TSMP
- KU13.** communication protocols and escalation procedures
- KU14.** basic project scheduling impacts on safety (sequence, overlap of works)



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret TSMP, method statements and statutory documents
- GS2.** prepare concise safety reports and inspection records
- GS3.** deliver clear safety briefings and toolbox talks
- GS4.** apply decision making to stop work or issue corrective actions
- GS5.** coordinate with multi-disciplinary teams and contractors
- GS6.** maintain accurate, legible records and registers
- GS7.** use digital tools for reporting and record keeping (if provided)
- GS8.** prioritise safety issues under time constraints
- GS9.** communicate effectively with workforce in local language(s)
- GS10.** basic numerical skills for recording measurements and logs
- GS11.** problem solving to close non-conformances
- GS12.** coaching and mentoring to improve worker competency
- GS13.** negotiation skills with contractors and suppliers on safety issues
- GS14.** observe and report unsafe acts objectively



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Establish Safety Management Arrangements for Tunnel Works</i>	6	12	-	2
PC1. confirm and implement the project Tunnel Safety Management Plan (TSMP) as issued by project management	-	-	-	-
PC2. ensure roles, responsibilities and communication lines for safety on the tunnelling site are documented and displayed	-	-	-	-
PC3. coordinate with project EHS and engineering teams to integrate TSMP with construction programme	-	-	-	-
PC4. verify that required permits, licenses and statutory registers are available on site	-	-	-	-
PC5. ensure safety briefings and induction schedules are planned and delivered for all incoming personnel	-	-	-	-
<i>Monitor Compliance with Safety Procedures and Statutory Requirements</i>	6	12	-	2
PC6. conduct routine site safety inspections and record non-conformances	-	-	-	-
PC7. ensure implementation of corrective and preventive actions within agreed timeframes	-	-	-	-
PC8. verify the use and condition of safety equipment and PPE as per TSMP	-	-	-	-
PC9. report incidents, near misses and unsafe acts/conditions as per reporting procedures	-	-	-	-
PC10. liaise with regulators and project QA/EHS for audits and inspections	-	-	-	-
<i>Safety Documentation and Record-Keeping</i>	6	12	-	2
PC11. maintain daily safety logs, gas monitoring records, ventilation logs and permits (confined space, hot work, LOTO, blasting)	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure accurate attendance, competency and medical fitness records for tunnel workforce	-	-	-	-
PC13. manage issuance, review and archival of method statements and risk assessments	-	-	-	-
PC14. prepare and submit periodic safety performance reports to site management.	-	-	-	-
PC15. ensure signage, hazard maps and emergency contact lists are current and accessible	-	-	-	-
<i>Training, Competency and Toolbox Talks</i>	6	12	-	2
PC16. identify safety training needs and ensure competence verification before assignment	-	-	-	-
PC17. schedule and conduct site-specific toolbox talks and safety drill	-	-	-	-
PC18. ensure specialist training (confined space, breathing apparatus, gas monitoring, emergency rescue) is current	-	-	-	-
PC19. monitor on-the-job coaching and corrective instruction for unsafe behaviour	-	-	-	-
PC20. maintain training matrix and certificates for audit	-	-	-	-
<i>Contractor/Subcontractor Safety Management</i>	6	12	-	2
PC21. verify that subcontractors submit safety method statements and risk assessments before commencement	-	-	-	-
PC22. conduct pre-start safety meetings with subcontractor supervisors	-	-	-	-
PC23. monitor subcontractor adherence to TSMP and take action on non-compliance	-	-	-	-
PC24. ensure coordination of interfaces (traffic, hoisting, muck handling) to prevent cross-contractor hazards	-	-	-	-
PC25. enforce contractor performance metrics and include safety KPIs in reviews	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	60	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N0601
NOS Name	Implement Tunnel Site Safety Management System
Sector	Construction
Sub-Sector	
Occupation	Safety works
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ICE/CON/N0602: Conduct Hazard Identification, Risk Assessment & Control for Tunnelling

Description

This NOS describes how to carry out continual hazard identification, formal risk assessments and implement control measures for tunnelling activities and changes in ground or work conditions.

Scope

The scope covers the following :

- Prepare for Hazard Identification and Risk Assessment
- Identify Hazards and Evaluate Risks
- Implement Risk Controls and Monitor Effectiveness
- Verify Competence for High-Risk Tasks
- Contractor/Subcontractor Safety Management

Elements and Performance Criteria

Prepare for Hazard Identification and Risk Assessment

To be competent, the user/individual on the job must be able to:

- PC1.** obtain available geotechnical, design and method statement documents
- PC2.** identify work activities, access routes and interfaces to be assessed
- PC3.** assemble competent assessment team including ground/geo engineers and equipment supervisors
- PC4.** select and confirm appropriate risk assessment method (JSA, HAZID, HAZOP, SWMS)
- PC5.** schedule assessments to coincide with high-risk activities and changes in conditions

Identify Hazards and Evaluate Risks

To be competent, the user/individual on the job must be able to:

- PC6.** identify hazards related to ground instability, rockfall, water ingress, gas, dust, ventilation, machinery, lifting and blasting
- PC7.** assess likelihood and consequence to determine risk rating using agreed matrix
- PC8.** record existing controls and their effectiveness
- PC9.** consider cumulative and emergent risks from work sequencing and environmental changes
- PC10.** communicate assessment findings to site management and affected teams

Implement Risk Controls and Monitor Effectiveness

To be competent, the user/individual on the job must be able to:

- PC11.** define and implement hierarchy of controls: elimination, substitution, engineering, administrative and PPE
- PC12.** ensure appropriate ground support, shoring, mesh, rock bolting, and shotcrete measures are implemented where identified
- PC13.** implement temporary access, exclusion zones and signage for hazards
- PC14.** assign responsible persons for implementation and monitoring of controls



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PC15. review controls after incidents, near misses, or condition changes and update RA/SWMS

Verify Competence for High-Risk Tasks

To be competent, the user/individual on the job must be able to:

PC16. establish procedure for immediate stop and reassessment when ground conditions change

PC17. ensure geotechnical and design team input for changed conditions

PC18. update method statements and risk assessments promptly and re-communicate to workforce

PC19. control and supervise any trial or remedial works

PC20. record decisions and approvals for change control

Contractor/Subcontractor Safety Management

To be competent, the user/individual on the job must be able to:

PC21. ensure persons undertaking tasks (bolting, scaling, blasting, TBM operation) are competent and certified where required

PC22. verify PPE, tools and equipment suitability for the identified hazard controls

PC23. supervise high-risk activities and ensure spot checks are performed

PC24. maintain competency and certification records for high-risk roles

PC25. conduct post-task reviews to capture lessons and update controls

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. principles of hazard identification and formal risk assessment methodologies

KU2. typical tunnelling hazards (ground collapse, rockfall, inundation, toxic gases, dust, noise, vibration)

KU3. hierarchy of controls and practical engineering control measures for tunnels

KU4. methods of ground support: rock bolting, shotcrete, steel ribs, mesh and face support

KU5. indicators of unstable ground and signs to stop work

KU6. interaction between construction methods (TBM, drill & blast, NATM) and hazard profiles

KU7. permit & authorization requirements for high-risk activities

KU8. role of geotechnical monitoring (instruments: extensometers, convergence pins, piezometers)

KU9. emergency stop, isolation and evacuation procedures

KU10. legal responsibilities of supervisors in hazard control

KU11. behavioural safety and near-miss reporting importance

KU12. control of hazardous substances (e.g., diesel exhaust, silica)

KU13. communication and signposting requirements for exclusion zones

KU14. methods for reviewing and updating risk assessments

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** perform structured risk assessments and document findings
- GS2.** read and interpret geotechnical reports and monitoring data at a supervisory level
- GS3.** communicate risk outcomes and control measures to workers and management
- GS4.** monitor implementation of engineering controls and enforcement of exclusion zones
- GS5.** apply judgement to halt operations when risk is unacceptable
- GS6.** record keeping and updating of RA/SWMS documentation
- GS7.** facilitate multidisciplinary risk workshops
- GS8.** coaching workers on safe work methods
- GS9.** analytical skills to interpret instrument readings and trends
- GS10.** time management to prioritise high-risk tasks
- GS11.** report writing for incident and risk review documentation
- GS12.** use of basic software tools for risk registers (if available)
- GS13.** conflict resolution with contractors when safety controls impact productivity



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for Hazard Identification and Risk Assessment</i>	6	12	-	2
PC1. obtain available geotechnical, design and method statement documents	-	-	-	-
PC2. identify work activities, access routes and interfaces to be assessed	-	-	-	-
PC3. assemble competent assessment team including ground/geo engineers and equipment supervisors	-	-	-	-
PC4. select and confirm appropriate risk assessment method (JSA, HAZID, HAZOP, SWMS)	-	-	-	-
PC5. schedule assessments to coincide with high-risk activities and changes in conditions	-	-	-	-
<i>Identify Hazards and Evaluate Risks</i>	6	12	-	2
PC6. identify hazards related to ground instability, rockfall, water ingress, gas, dust, ventilation, machinery, lifting and blasting	-	-	-	-
PC7. assess likelihood and consequence to determine risk rating using agreed matrix	-	-	-	-
PC8. record existing controls and their effectiveness	-	-	-	-
PC9. consider cumulative and emergent risks from work sequencing and environmental changes	-	-	-	-
PC10. communicate assessment findings to site management and affected teams	-	-	-	-
<i>Implement Risk Controls and Monitor Effectiveness</i>	6	12	-	2
PC11. define and implement hierarchy of controls: elimination, substitution, engineering, administrative and PPE	-	-	-	-
PC12. ensure appropriate ground support, shoring, mesh, rock bolting, and shotcrete measures are implemented where identified	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. implement temporary access, exclusion zones and signage for hazards	-	-	-	-
PC14. assign responsible persons for implementation and monitoring of controls	-	-	-	-
PC15. review controls after incidents, near misses, or condition changes and update RA/SWMS	-	-	-	-
<i>Verify Competence for High-Risk Tasks</i>	6	12	-	2
PC16. establish procedure for immediate stop and reassessment when ground conditions change	-	-	-	-
PC17. ensure geotechnical and design team input for changed conditions	-	-	-	-
PC18. update method statements and risk assessments promptly and re-communicate to workforce	-	-	-	-
PC19. control and supervise any trial or remedial works	-	-	-	-
PC20. record decisions and approvals for change control	-	-	-	-
<i>Contractor/Subcontractor Safety Management</i>	6	12	-	2
PC21. ensure persons undertaking tasks (bolting, scaling, blasting, TBM operation) are competent and certified where required	-	-	-	-
PC22. verify PPE, tools and equipment suitability for the identified hazard controls	-	-	-	-
PC23. supervise high-risk activities and ensure spot checks are performed	-	-	-	-
PC24. maintain competency and certification records for high-risk roles	-	-	-	-
PC25. conduct post-task reviews to capture lessons and update controls	-	-	-	-
NOS Total	30	60	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N0602
NOS Name	Conduct Hazard Identification, Risk Assessment & Control for Tunnelling
Sector	Construction
Sub-Sector	
Occupation	Safety works
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ICE/CON/N0603: Manage Ventilation, Gas Monitoring and Occupational Hygiene in Tunnels

Description

This NOS describes how to supervise and manage tunnel ventilation systems, continuous gas monitoring, and occupational hygiene controls to maintain safe breathable atmosphere and acceptable exposure limits.

Scope

The scope covers the following :

- Plan Ventilation and Monitoring Needs
- Implement and Monitor Gas and Air Quality Controls
- Maintain and Test Ventilation and Monitoring Equipment
- Occupational Hygiene Controls and Worker Protection
- Response to Gas/Air Quality Alarms and Incidents

Elements and Performance Criteria

Plan Ventilation and Monitoring Needs

To be competent, the user/individual on the job must be able to:

- PC1.** review design ventilation requirements and method statements for activity types
- PC2.** identify location and type of fixed and portable gas monitors and air sampling points
- PC3.** ensure ventilation equipment (fans, ducts, silencers) is available, certified and correctly sized
- PC4.** confirm maintenance and calibration schedules for monitors and fans
- PC5.** coordinate with plant operators to establish airflow routes and contingencies

Implement and Monitor Gas and Air Quality Controls

To be competent, the user/individual on the job must be able to:

- PC6.** ensure continuous monitoring for oxygen, carbon monoxide, methane, NOx and other specified gases
- PC7.** record and interpret gas readings and take immediate action on exceedances (evacuation, ventilation increase)
- PC8.** monitor particulate (dust) levels and ensure dust suppression measures are in place (water sprays, extraction)
- PC9.** ensure use of respiratory protective equipment when controls cannot adequately reduce exposures
- PC10.** ensure isolation of ignition sources if flammable gas detected and follow hot-work permit suspension

Maintain and Test Ventilation and Monitoring Equipment

To be competent, the user/individual on the job must be able to:

- PC11.** ensure pre-shift and periodic checks on fans, ducting, filters and monitors are carried out and recorded



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- PC12.** ensure routine bump tests and calibrations of portable gas detectors are completed
- PC13.** arrange preventive maintenance and rapid repairs to ventilation plant to prevent loss of airflow
- PC14.** keep logs of airflow measurements and calibration certificates
- PC15.** implement redundancy plans for critical plant failures (spare fans, backup power)

Occupational Hygiene Controls and Worker Protection

To be competent, the user/individual on the job must be able to:

- PC16.** ensure health surveillance, baseline medicals and occupational hygiene monitoring as required
- PC17.** confirm controls for noise, vibration, diesel exhaust and silica dust are implemented
- PC18.** ensure fit testing of respirators and that RPE selection matches hazard
- PC19.** coordinate with health team for exposure records and corrective measures
- PC20.** promote hygiene facilities (washing, separate eating areas) to reduce ingestion/contamination risks

Response to Gas/Air Quality Alarms and Incidents

To be competent, the user/individual on the job must be able to:

- PC21.** implement emergency ventilation and evacuation procedures on alarm
- PC22.** ensure rescue and medical arrangements for contaminated atmosphere events
- PC23.** preserve and hand over monitoring logs and instrument data for investigation
- PC24.** debrief and update control measures following an air quality incident
- PC25.** ensure corrective actions to prevent recurrence are tracked and closed

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles of tunnel ventilation (supply/exhaust, longitudinal/transverse flow)
- KU2.** typical hazardous gases in tunnelling and their health/effective concentration limits
- KU3.** use, calibration and limitations of portable and fixed gas detectors
- KU4.** occupational exposure limits for dust, diesel particulate, silica, noise and vibration
- KU5.** selection, use and maintenance of respiratory protective equipment and fit testing
- KU6.** methods of dust suppression and extraction in confined spaces
- KU7.** interaction between ventilation, temperature and worker comfort
- KU8.** emergency ventilation strategies and safe evacuation procedures
- KU9.** instrumentation logs, data handling and record retention requirements
- KU10.** health surveillance obligations and medical fitness requirements
- KU11.** electrical & ignition source controls in potentially flammable atmospheres
- KU12.** requirements for redundancy and spare parts for critical ventilation systems
- KU13.** legal and project requirements for atmospheric monitoring

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** interpret gas readings and decide immediate control actions
- GS2.** maintain calibration and maintenance records with attention to detail
- GS3.** communicate air quality issues clearly to workforce and management
- GS4.** coordinate with maintenance teams to prioritise ventilation repairs
- GS5.** conduct basic airflow measurement checks and logging
- GS6.** organise and supervise fit testing and RPE issuance
- GS7.** prepare concise incident summaries related to atmospheres
- GS8.** prioritise tasks when multiple monitoring alarms occur
- GS9.** use instruments and troubleshoot common faults
- GS10.** persuade and enforce use of hygiene and RPE controls
- GS11.** data-driven decision making for ventilation adjustments
- GS12.** liaise with health professionals for exposure management



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plan Ventilation and Monitoring Needs</i>	6	12	-	2
PC1. review design ventilation requirements and method statements for activity types	-	-	-	-
PC2. identify location and type of fixed and portable gas monitors and air sampling points	-	-	-	-
PC3. ensure ventilation equipment (fans, ducts, silencers) is available, certified and correctly sized	-	-	-	-
PC4. confirm maintenance and calibration schedules for monitors and fans	-	-	-	-
PC5. coordinate with plant operators to establish airflow routes and contingencies	-	-	-	-
<i>Implement and Monitor Gas and Air Quality Controls</i>	6	12	-	2
PC6. ensure continuous monitoring for oxygen, carbon monoxide, methane, NOx and other specified gases	-	-	-	-
PC7. record and interpret gas readings and take immediate action on exceedances (evacuation, ventilation increase)	-	-	-	-
PC8. monitor particulate (dust) levels and ensure dust suppression measures are in place (water sprays, extraction)	-	-	-	-
PC9. ensure use of respiratory protective equipment when controls cannot adequately reduce exposures	-	-	-	-
PC10. ensure isolation of ignition sources if flammable gas detected and follow hot-work permit suspension	-	-	-	-
<i>Maintain and Test Ventilation and Monitoring Equipment</i>	6	12	-	2
PC11. ensure pre-shift and periodic checks on fans, ducting, filters and monitors are carried out and recorded	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure routine bump tests and calibrations of portable gas detectors are completed	-	-	-	-
PC13. arrange preventive maintenance and rapid repairs to ventilation plant to prevent loss of airflow	-	-	-	-
PC14. keep logs of airflow measurements and calibration certificates	-	-	-	-
PC15. implement redundancy plans for critical plant failures (spare fans, backup power)	-	-	-	-
<i>Occupational Hygiene Controls and Worker Protection</i>	6	12	-	2
PC16. ensure health surveillance, baseline medicals and occupational hygiene monitoring as required	-	-	-	-
PC17. confirm controls for noise, vibration, diesel exhaust and silica dust are implemented	-	-	-	-
PC18. ensure fit testing of respirators and that RPE selection matches hazard	-	-	-	-
PC19. coordinate with health team for exposure records and corrective measures	-	-	-	-
PC20. promote hygiene facilities (washing, separate eating areas) to reduce ingestion/contamination risks	-	-	-	-
<i>Response to Gas/Air Quality Alarms and Incidents</i>	6	12	-	2
PC21. implement emergency ventilation and evacuation procedures on alarm	-	-	-	-
PC22. ensure rescue and medical arrangements for contaminated atmosphere events	-	-	-	-
PC23. preserve and hand over monitoring logs and instrument data for investigation	-	-	-	-
PC24. debrief and update control measures following an air quality incident	-	-	-	-
PC25. ensure corrective actions to prevent recurrence are tracked and closed	-	-	-	-
NOS Total	30	60	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N0603
NOS Name	Manage Ventilation, Gas Monitoring and Occupational Hygiene in Tunnels
Sector	Construction
Sub-Sector	
Occupation	Safety works
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ICE/CON/N0604: Emergency Preparedness, Rescue & Evacuation in Tunnel Works

Description

This NOS describes how to plan, implement and lead emergency preparedness, rescue and evacuation systems specific to tunnelling environments including confined space rescue, fire, inundation and collapse scenarios.

Scope

The scope covers the following :

- Develop and Maintain Tunnel Emergency Plans
- Establish Rescue Team and Training
- Emergency Detection, Alarm and Communication
- Execute Evacuation and Rescue Operations
- Review, Learn and Improve Emergency Arrangements

Elements and Performance Criteria

Develop and Maintain Tunnel Emergency Plans

To be competent, the user/individual on the job must be able to:

- PC1.** prepare site-specific emergency response plans for foreseeable tunnel incidents (fire, gas leak, collapse, flooding)
- PC2.** ensure plans include roles, communication lines, muster points and rescue resources
- PC3.** coordinate with local emergency services and project emergency coordination team
- PC4.** ensure availability and location mapping of rescue equipment (breathing apparatus, stretchers, rescue harnesses, ventilation spares)
- PC5.** keep up-to-date contact lists and incident decision trees

Establish Rescue Team and Training

To be competent, the user/individual on the job must be able to:

- PC6.** identify, train and maintain a competent on-site rescue team for confined space and tunnel rescue
- PC7.** ensure team members have valid certificates for breathing apparatus, rope rescue, first aid and casualty handling
- PC8.** schedule routine drills (full scale and tabletop) and record outcomes and improvement actions
- PC9.** ensure rescue team is rostered and available for shiftwork and emergency call-outs
- PC10.** provide refresher training and competency assessment on rescue procedures

Emergency Detection, Alarm and Communication

To be competent, the user/individual on the job must be able to:

- PC11.** ensure reliable alarm/PA systems are installed and tested



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- PC12.** confirm redundant communication methods (cable phones, UHF radios, leaky feeder, runner) are functional in tunnel zones
- PC13.** ensure clear procedures for raising, escalating and recording alarms
- PC14.** test failover communications during drills and maintain logs
- PC15.** ensure emergency signage and escape wayfinding are visible and maintained

Execute Evacuation and Rescue Operations

To be competent, the user/individual on the job must be able to:

- PC16.** confirm situation, isolate hazard (if safe) and initiate evacuation plan on receipt of alarm
- PC17.** coordinate rescue team deployment, ensuring their safety (breathing apparatus, airway management)
- PC18.** manage casualty handling and handover to onsite medical facility or external services
- PC19.** secure scene for investigation after immediate rescue and medical needs are met
- PC20.** initiate post-incident welfare support and worker account for all personnel

Review, Learn and Improve Emergency Arrangements

To be competent, the user/individual on the job must be able to:

- PC21.** conduct after-action reviews and incident investigations with multidisciplinary inputs
- PC22.** update emergency plans, training records and control measures based on lessons learned
- PC23.** maintain inventory, condition and service records of rescue equipment
- PC24.** ensure corrective actions from drills or incidents are implemented and tracked
- PC25.** communicate improvements and revised plans to all site personnel

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of tunnel emergencies and typical causes (fire, gas, collapse, flooding)
- KU2.** principles of confined space rescue and breathing apparatus operation
- KU3.** requirements for rescue team composition, training and certification
- KU4.** communication systems used in tunnels and their redundancy requirements
- KU5.** legal and project obligations for emergency preparedness and drills
- KU6.** first aid and casualty handling specific to tunnel injuries (asphyxiation, crush, burns, hypothermia)
- KU7.** rescue equipment types, maintenance and readiness checks
- KU8.** procedures for scene preservation and incident investigation
- KU9.** psychological first aid and post-incident welfare requirements
- KU10.** coordination with external emergency services and handover protocols
- KU11.** logging and reporting requirements after an emergency
- KU12.** establishing and maintaining muster points and escape routes

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** lead emergency drills and evaluate team performance
- GS2.** make rapid decisions under pressure safeguarding rescuers and casualties
- GS3.** coordinate multi-agency responses and liaise with external services
- GS4.** conduct post-incident debriefs and document action items
- GS5.** maintain rescue equipment records and readiness checks
- GS6.** communicate instructions clearly to panicked or injured workers
- GS7.** train and mentor rescue team members
- GS8.** apply incident command principles and maintain situational awareness
- GS9.** analyse drill outcomes and implement improvements
- GS10.** maintain calm and provide leadership during evacuations
- GS11.** plan logistics for emergency response resources
- GS12.** keep accurate logs for legal and insurance purposes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Develop and Maintain Tunnel Emergency Plans</i>	6	12	-	2
PC1. prepare site-specific emergency response plans for foreseeable tunnel incidents (fire, gas leak, collapse, flooding)	-	-	-	-
PC2. ensure plans include roles, communication lines, muster points and rescue resources	-	-	-	-
PC3. coordinate with local emergency services and project emergency coordination team	-	-	-	-
PC4. ensure availability and location mapping of rescue equipment (breathing apparatus, stretchers, rescue harnesses, ventilation spares)	-	-	-	-
PC5. keep up-to-date contact lists and incident decision trees	-	-	-	-
<i>Establish Rescue Team and Training</i>	6	12	-	2
PC6. identify, train and maintain a competent on-site rescue team for confined space and tunnel rescue	-	-	-	-
PC7. ensure team members have valid certificates for breathing apparatus, rope rescue, first aid and casualty handling	-	-	-	-
PC8. schedule routine drills (full scale and tabletop) and record outcomes and improvement actions	-	-	-	-
PC9. ensure rescue team is rostered and available for shiftwork and emergency call-outs	-	-	-	-
PC10. provide refresher training and competency assessment on rescue procedures	-	-	-	-
<i>Emergency Detection, Alarm and Communication</i>	6	12	-	2
PC11. ensure reliable alarm/PA systems are installed and tested	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. confirm redundant communication methods (cable phones, UHF radios, leaky feeder, runner) are functional in tunnel zones	-	-	-	-
PC13. ensure clear procedures for raising, escalating and recording alarms	-	-	-	-
PC14. test failover communications during drills and maintain logs	-	-	-	-
PC15. ensure emergency signage and escape wayfinding are visible and maintained	-	-	-	-
<i>Execute Evacuation and Rescue Operations</i>	6	12	-	2
PC16. confirm situation, isolate hazard (if safe) and initiate evacuation plan on receipt of alarm	-	-	-	-
PC17. coordinate rescue team deployment, ensuring their safety (breathing apparatus, airway management)	-	-	-	-
PC18. manage casualty handling and handover to onsite medical facility or external services	-	-	-	-
PC19. secure scene for investigation after immediate rescue and medical needs are met	-	-	-	-
PC20. initiate post-incident welfare support and worker account for all personnel	-	-	-	-
<i>Review, Learn and Improve Emergency Arrangements</i>	6	12	-	2
PC21. conduct after-action reviews and incident investigations with multidisciplinary inputs	-	-	-	-
PC22. update emergency plans, training records and control measures based on lessons learned	-	-	-	-
PC23. maintain inventory, condition and service records of rescue equipment	-	-	-	-
PC24. ensure corrective actions from drills or incidents are implemented and tracked	-	-	-	-
PC25. communicate improvements and revised plans to all site personnel	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N0604
NOS Name	Emergency Preparedness, Rescue & Evacuation in Tunnel Works
Sector	Construction
Sub-Sector	
Occupation	Safety works
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N0605: Manage Ground Support, Excavation Face Safety and Rockfall Controls

Description

This NOS describes how to supervise and ensure safe execution of ground support, excavation face operations, scaling, rock bolting, shotcrete, and temporary works for tunnel stability.

Scope

The scope covers the following :

- Prepare Ground Support Plan Implementation
- Supervise Rock Bolting and Mesh Installation
- Supervise Shotcrete and Temporary Lining Works
- Manage Testing, Instrumentation and Monitoring
- Control Working at Face and Exclusion Zones

Elements and Performance Criteria

Prepare Ground Support Plan Implementation

To be competent, the user/individual on the job must be able to:

- PC1.** review ground support design and manufacturer's instructions for support materials
- PC2.** confirm availability and quality certificates of support materials (bolts, mesh, shotcrete)
- PC3.** ensure method statements specify installation sequence, equipment and QA checks
- PC4.** establish exclusion zones and access control for face works
- PC5.** ensure scaffoldings, platforms and anchorage for installers comply with design

Supervise Rock Bolting and Mesh Installation

To be competent, the user/individual on the job must be able to:

- PC6.** verify drilling patterns, bolt lengths and torque/anchorage as per design
- PC7.** ensure proper scaling and removal of loose material before support installation
- PC8.** confirm installation checks (pull tests, torque checks) are performed and recorded
- PC9.** enforce safe working distances and PPE for installers
- PC10.** ensure splicing/overlapping of mesh and correct fixation to anchors

Supervise Shotcrete and Temporary Lining Works

To be competent, the user/individual on the job must be able to:

- PC11.** ensure shotcrete mix, delivery and nozzle operator procedures meet specified design
- PC12.** confirm curing, reinforcement placement and lining thickness checks
- PC13.** implement controls for rebound, dust suppression and nozzle safety
- PC14.** ensure inspection of temporary lining before access is permitted
- PC15.** coordinate with ventilation to control dust and exhaust when shotcreting

Manage Testing, Instrumentation and Monitoring

To be competent, the user/individual on the job must be able to:



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- PC16.** ensure geotechnical instrumentation is installed and readings logged (convergence, load on bolts)
- PC17.** review instrumentation data and escalate anomalies
- PC18.** schedule and supervise pre-task and post-task inspections of face and support
- PC19.** ensure corrective measures are implemented for observed instability
- PC20.** maintain a record of tests, inspections and remedial actions

Control Working at Face and Exclusion Zones

To be competent, the user/individual on the job must be able to:

- PC21.** enforce single-entry procedures and access permits for face areas
- PC22.** monitor personnel movement and ensure no-go zones are respected during high-risk activities
- PC23.** manage mucking and haulage to avoid undermining face support or creating vibration
- PC24.** ensure equipment used near face is correctly rated and inspected
- PC25.** stop operations when unsafe conditions are observed and re-assess

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** ground support methods and applications (rock bolting, mesh, shotcrete, steel ribs)
- KU2.** excavation face hazards and mitigation measures
- KU3.** requirements for material quality, testing and manufacturer's instructions
- KU4.** instrumentation types (convergence, extensometers) and interpretation basics
- KU5.** installation procedures for rock bolts and shotcrete
- KU6.** access control, exclusion zones and safe distances for face works
- KU7.** interaction between blasting/vibration and ground support performance
- KU8.** temporary works and scaffolding safety near face
- KU9.** dust control, rebound and nozzle safety during shotcreting
- KU10.** permit and inspection protocols for face access
- KU11.** legal obligations and industry good practice for ground control
- KU12.** emergency actions for sudden face instability

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and apply ground support design drawings and method statements
- GS2.** supervise and confirm installation quality (visual checks and records)
- GS3.** interpret basic instrument outputs and escalate issues
- GS4.** coordinate activities (bolting, shotcrete, mucking) to minimise conflict
- GS5.** enforce access permit systems and exclusion zones
- GS6.** produce concise installation and inspection records



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- GS7.** problem solve when unexpected ground conditions appear
- GS8.** coach installers on correct techniques and safety precautions
- GS9.** plan logistics for materials, plant and manpower for support works
- GS10.** use hand-held tools and confirm their safe condition
- GS11.** communicate clearly with geotechnical engineers and designers
- GS12.** prioritise remedial actions when instability is detected



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare Ground Support Plan Implementation</i>	6	12	-	2
PC1. review ground support design and manufacturer's instructions for support materials	-	-	-	-
PC2. confirm availability and quality certificates of support materials (bolts, mesh, shotcrete)	-	-	-	-
PC3. ensure method statements specify installation sequence, equipment and QA checks	-	-	-	-
PC4. establish exclusion zones and access control for face works	-	-	-	-
PC5. ensure scaffoldings, platforms and anchorage for installers comply with design	-	-	-	-
<i>Supervise Rock Bolting and Mesh Installation</i>	6	12	-	2
PC6. verify drilling patterns, bolt lengths and torque/anchorage as per design	-	-	-	-
PC7. ensure proper scaling and removal of loose material before support installation	-	-	-	-
PC8. confirm installation checks (pull tests, torque checks) are performed and recorded	-	-	-	-
PC9. enforce safe working distances and PPE for installers	-	-	-	-
PC10. ensure splicing/overlapping of mesh and correct fixation to anchors	-	-	-	-
<i>Supervise Shotcrete and Temporary Lining Works</i>	6	12	-	2
PC11. ensure shotcrete mix, delivery and nozzle operator procedures meet specified design	-	-	-	-
PC12. confirm curing, reinforcement placement and lining thickness checks	-	-	-	-
PC13. implement controls for rebound, dust suppression and nozzle safety	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure inspection of temporary lining before access is permitted	-	-	-	-
PC15. coordinate with ventilation to control dust and exhaust when shotcreting	-	-	-	-
<i>Manage Testing, Instrumentation and Monitoring</i>	6	12	-	2
PC16. ensure geotechnical instrumentation is installed and readings logged (convergence, load on bolts)	-	-	-	-
PC17. review instrumentation data and escalate anomalies	-	-	-	-
PC18. schedule and supervise pre-task and post-task inspections of face and support	-	-	-	-
PC19. ensure corrective measures are implemented for observed instability	-	-	-	-
PC20. maintain a record of tests, inspections and remedial actions	-	-	-	-
<i>Control Working at Face and Exclusion Zones</i>	6	12	-	2
PC21. enforce single-entry procedures and access permits for face areas	-	-	-	-
PC22. monitor personnel movement and ensure no-go zones are respected during high-risk activities	-	-	-	-
PC23. manage mucking and haulage to avoid undermining face support or creating vibration	-	-	-	-
PC24. ensure equipment used near face is correctly rated and inspected	-	-	-	-
PC25. stop operations when unsafe conditions are observed and re-assess	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N0605
NOS Name	Manage Ground Support, Excavation Face Safety and Rockfall Controls
Sector	Construction
Sub-Sector	
Occupation	Safety works
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N0606: Manage Plant, TBM, Mechanical & Blasting Safety in Tunnelling

Description

This NOS describes how to supervise safe operation of tunnelling plant and equipment including TBMs, drilling rigs, loaders, conveyors, and manage blasting & explosives safety where applicable.

Scope

The scope covers the following :

- Ensure Plant/Equipment Safety and Certification
- Supervise TBM and Mechanised Operations
- Manage Drill & Blast Operations
- Control Mobile Plant Interactions and Haulage
- Manage LOTO, Maintenance and Emergency Stop Systems

Elements and Performance Criteria

Ensure Plant/Equipment Safety and Certification

To be competent, the user/individual on the job must be able to:

- PC1.** verify plant has valid inspection and certification and is suitable for the intended tunnelling task
- PC2.** ensure pre-shift and periodic plant checks are completed and logged
- PC3.** confirm guarding, interlocks and emergency stops are functional
- PC4.** ensure operators hold required licences/certificates and undergo site induction
- PC5.** manage fuel, lubrication and hazardous substances storage in line with safety procedures

Supervise TBM and Mechanised Operations

To be competent, the user/individual on the job must be able to:

- PC6.** confirm TBM safety systems (overpressure protection, egress, cutter inspection) are maintained
- PC7.** ensure TBM work procedures (cutter change, segment installation) are carried out under safe conditions and method statements
- PC8.** coordinate TBM thrust, backup activities and muck removal to maintain stability and safe access
- PC9.** ensure confined space and egress procedures for TBM interior work
- PC10.** verify lockout/tagout and control measures during TBM maintenance

Manage Drill & Blast Operations

To be competent, the user/individual on the job must be able to:

- PC11.** ensure explosives handling, storage, transport and use comply with statutory requirements and project procedures
- PC12.** ensure blasting plans including delay patterns, stemming and exclusion zones are reviewed and authorized



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PC13. verify pre-blast warnings, evacuation and monitoring are completed

PC14. ensure post-blast inspection and clearance before re-entry

PC15. maintain explosives register and competency records for blasters

Control Mobile Plant Interactions and Haulage

To be competent, the user/individual on the job must be able to:

PC16. ensure segregation of pedestrian and mobile plant routes in tunnel work areas

PC17. implement vehicle short-loading, reversing and illumination controls in confined spaces

PC18. verify maintenance regimes and emergency response for mobile plant incidents

PC19. ensure drivers/operators adhere to speed limits and route controls

PC20. supervise safe coupling/uncoupling of muck haulage equipment

Manage LOTO, Maintenance and Emergency Stop Systems

To be competent, the user/individual on the job must be able to:

PC21. ensure LOTO procedures are in place and applied for mechanical and electrical maintenance

PC22. verify lockout devices, permits and isolation records before maintenance

PC23. ensure emergency stop and rescue access routes around plant are maintained

PC24. ensure competence and supervision during plant repairs and align with manufacturer instructions

PC25. record and investigate plant-related incidents to prevent recurrence

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. plant inspection and certification requirements for tunnelling equipment

KU2. TBM design basics and safety systems (egress, cutterhead safety)

KU3. statutory explosives and blasting regulations and safe handling practices

KU4. LOTO procedures, electrical isolation and high-voltage safety basics

KU5. mobile plant operation hazards in confined spaces (visibility, ventilation)

KU6. safe procedures for working adjacent to moving plant and conveyors

KU7. fuel and hazardous chemical storage and spill response

KU8. requirements for operator licensing and competency

KU9. emergency stop systems and mechanical fail-safe principles

KU10. reporting and investigation of plant incidents

KU11. maintenance scheduling for critical plant and spare parts management

KU12. TBM backup system safety and cutter replacement procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. review plant certificates and maintenance logs for currency

GS2. supervise operator competency checks and licensing



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- GS3.** coordinate TBM operations with construction sequencing and safety teams
- GS4.** apply LOTO processes and verify tagout completeness
- GS5.** communicate and enforce vehicle segregation and traffic controls
- GS6.** analyse plant incident causes and recommend corrective actions
- GS7.** maintain explosives registers and ensure compliance checks
- GS8.** plan maintenance windows with minimal safety impact
- GS9.** ensure safe stores and handling of fuels and hazardous chemicals
- GS10.** use checklists for pre-start and post-maintenance inspections
- GS11.** organise toolbox talks focused on plant related hazards
- GS12.** liaise with equipment manufacturer/service contractors for complex repairs



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure Plant/Equipment Safety and Certification</i>	6	12	-	2
PC1. verify plant has valid inspection and certification and is suitable for the intended tunnelling task	-	-	-	-
PC2. ensure pre-shift and periodic plant checks are completed and logged	-	-	-	-
PC3. confirm guarding, interlocks and emergency stops are functional	-	-	-	-
PC4. ensure operators hold required licences/certificates and undergo site induction	-	-	-	-
PC5. manage fuel, lubrication and hazardous substances storage in line with safety procedures	-	-	-	-
<i>Supervise TBM and Mechanised Operations</i>	6	12	-	2
PC6. confirm TBM safety systems (overpressure protection, egress, cutter inspection) are maintained	-	-	-	-
PC7. ensure TBM work procedures (cutter change, segment installation) are carried out under safe conditions and method statements	-	-	-	-
PC8. coordinate TBM thrust, backup activities and muck removal to maintain stability and safe access	-	-	-	-
PC9. ensure confined space and egress procedures for TBM interior work	-	-	-	-
PC10. verify lockout/tagout and control measures during TBM maintenance	-	-	-	-
<i>Manage Drill & Blast Operations</i>	6	12	-	2
PC11. ensure explosives handling, storage, transport and use comply with statutory requirements and project procedures	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure blasting plans including delay patterns, stemming and exclusion zones are reviewed and authorized	-	-	-	-
PC13. verify pre-blast warnings, evacuation and monitoring are completed	-	-	-	-
PC14. ensure post-blast inspection and clearance before re-entry	-	-	-	-
PC15. maintain explosives register and competency records for blasters	-	-	-	-
<i>Control Mobile Plant Interactions and Haulage</i>	6	12	-	2
PC16. ensure segregation of pedestrian and mobile plant routes in tunnel work areas	-	-	-	-
PC17. implement vehicle short-loading, reversing and illumination controls in confined spaces	-	-	-	-
PC18. verify maintenance regimes and emergency response for mobile plant incidents	-	-	-	-
PC19. ensure drivers/operators adhere to speed limits and route controls	-	-	-	-
PC20. supervise safe coupling/uncoupling of muck haulage equipment	-	-	-	-
<i>Manage LOTO, Maintenance and Emergency Stop Systems</i>	6	12	-	2
PC21. ensure LOTO procedures are in place and applied for mechanical and electrical maintenance	-	-	-	-
PC22. verify lockout devices, permits and isolation records before maintenance	-	-	-	-
PC23. ensure emergency stop and rescue access routes around plant are maintained	-	-	-	-
PC24. ensure competence and supervision during plant repairs and align with manufacturer instructions	-	-	-	-
PC25. record and investigate plant-related incidents to prevent recurrence	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N0606
NOS Name	Manage Plant, TBM, Mechanical & Blasting Safety in Tunnelling
Sector	Construction
Sub-Sector	
Occupation	Safety works
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N9901: Implement Safe Work Practices and Environmental Stewardship at Construction Sites

Description

This NOS focuses on enabling construction workers and supervisors to actively adopt safe work practices, promote environmental stewardship and ensure a hazard-free workplace through proactive risk management, sustainable resource usage and timely emergency response.

Scope

The scope covers the following :

- Conduct workplace safety inspections
- Apply safe operational practices
- Promote health, hygiene, and well-being
- Implement sustainability and environmental conservation measures
- Respond effectively to onsite emergencies

Elements and Performance Criteria

Conduct Workplace Safety Inspections

To be competent, the user/individual on the job must be able to:

- PC1.** identify unsafe conditions such as improper scaffolding, poor lighting, open edges or obstructed pathways
- PC2.** assess task-specific risks involving tools, machinery, and hazardous materials
- PC3.** document safety observations and communicate them to the safety lead or supervisor
- PC4.** verify that safety controls such as guard rails, signages and access controls are in place

Apply Safe Operational Practices

To be competent, the user/individual on the job must be able to:

- PC5.** use appropriate PPE and ensure correct fit before beginning any activity
- PC6.** follow safe operation procedures for cutting, lifting, mixing and power tools
- PC7.** implement ergonomic practices to reduce fatigue and prevent musculoskeletal injuries
- PC8.** maintain housekeeping standards by ensuring clean and clutter-free work areas

Promote Health, Hygiene and Well-Being

To be competent, the user/individual on the job must be able to:

- PC9.** follow personal hygiene practices such as handwashing, clean clothing and proper hydration
- PC10.** identify signs of health issues such as heat stress, dehydration and respiratory discomfort
- PC11.** ensure availability and safe usage of sanitation facilities, drinking water and resting zones
- PC12.** report workplace illnesses or physical discomfort to the assigned authority

Implement Sustainability and Environmental Conservation Measures

To be competent, the user/individual on the job must be able to:

- PC13.** follow procedures for safe storage and controlled use of chemicals, fuels and hazardous substances



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PC14. practice material conservation by preventing spillage, wastage and improper handling

PC15. undertake waste segregation for recyclable, non-recyclable and hazardous waste

PC16. support measures for dust suppression, noise control and energy/water conservation

Respond Effectively to Onsite Emergencies

To be competent, the user/individual on the job must be able to:

PC17. identify emergency alarms, muster points and evacuation routes

PC18. use basic firefighting equipment such as extinguishers, fire blankets and sand buckets where appropriate

PC19. assist in basic first-aid procedures including bleeding control, CPR and heat-stroke management

PC20. support emergency communication protocols and follow command instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. workplace safety policies, EHS manuals and site-specific procedures

KU2. roles of safety personnel, supervisors and emergency response teams

KU3. documentation formats for safety reporting and compliance

KU4. internal escalation protocols for hazardous conditions or incidents

KU5. legal obligations related to occupational safety and environmental protection

KU6. types of workplace hazards chemical, physical, electrical, ergonomic and environmental

KU7. safe handling techniques for construction materials, tools and machinery

KU8. principles of PPE selection, inspection and maintenance

KU9. basics of first aid, CPR and firefighting methods

KU10. environmental management practices including waste reduction and pollution control

KU11. measures for heat-stress prevention, hydration management and fatigue control

KU12. ergonomics and safe manual handling techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret safety signage, manuals and instructions

GS2. communicate safety findings and concerns clearly to team members

GS3. maintain basic documentation and reports for safety audits

GS4. use problem-solving skills to manage minor hazards or disruptions

GS5. collaborate effectively during drills, emergency situations and environmental initiatives

GS6. demonstrate awareness and discipline in following safe work practices consistently



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct Workplace Safety Inspections</i>	6	12	-	2
PC1. identify unsafe conditions such as improper scaffolding, poor lighting, open edges or obstructed pathways	-	-	-	-
PC2. assess task-specific risks involving tools, machinery, and hazardous materials	-	-	-	-
PC3. document safety observations and communicate them to the safety lead or supervisor	-	-	-	-
PC4. verify that safety controls such as guard rails, signages and access controls are in place	-	-	-	-
<i>Apply Safe Operational Practices</i>	6	12	-	2
PC5. use appropriate PPE and ensure correct fit before beginning any activity	-	-	-	-
PC6. follow safe operation procedures for cutting, lifting, mixing and power tools	-	-	-	-
PC7. implement ergonomic practices to reduce fatigue and prevent musculoskeletal injuries	-	-	-	-
PC8. maintain housekeeping standards by ensuring clean and clutter-free work areas	-	-	-	-
<i>Promote Health, Hygiene and Well-Being</i>	6	12	-	2
PC9. follow personal hygiene practices such as handwashing, clean clothing and proper hydration	-	-	-	-
PC10. identify signs of health issues such as heat stress, dehydration and respiratory discomfort	-	-	-	-
PC11. ensure availability and safe usage of sanitation facilities, drinking water and resting zones	-	-	-	-
PC12. report workplace illnesses or physical discomfort to the assigned authority	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implement Sustainability and Environmental Conservation Measures</i>	6	12	-	2
PC13. follow procedures for safe storage and controlled use of chemicals, fuels and hazardous substances	-	-	-	-
PC14. practice material conservation by preventing spillage, wastage and improper handling	-	-	-	-
PC15. undertake waste segregation for recyclable, non-recyclable and hazardous waste	-	-	-	-
PC16. support measures for dust suppression, noise control and energy/water conservation	-	-	-	-
<i>Respond Effectively to Onsite Emergencies</i>	6	12	-	2
PC17. identify emergency alarms, muster points and evacuation routes	-	-	-	-
PC18. use basic firefighting equipment such as extinguishers, fire blankets and sand buckets where appropriate	-	-	-	-
PC19. assist in basic first-aid procedures including bleeding control, CPR and heat-stroke management	-	-	-	-
PC20. support emergency communication protocols and follow command instructions	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N9901
NOS Name	Implement Safe Work Practices and Environmental Stewardship at Construction Sites
Sector	Construction
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Awarding Body. Each Performance Criteria (PC)/ Element will be assigned marks proportional to its importance in NOS. AB will also lay down proportion of marks for Theory and Practical Skills for each PC/ Element.
2. The assessment of the knowledge part will be based on knowledge bank of questions created by Assessment Bodies subject to approval by AB.
3. Individual assessment agencies will create unique question papers for knowledge/theory part for assessment of candidates as per assessment criteria given below.
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on assessment criteria.
5. To pass the Qualification Pack, every trainee must score 70% on overall QP.
6. The Assessor shall check the outcome of the practices while evaluating the steps performed to achieve the outcome.
7. The trainee shall be provided with a chance to repeat the test to correct his procedures in case of



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improper performance, with a deduction of marks for each iteration.

8. After the certain number of iterations as decided by AB the trainee is marked as fail, scoring zero marks for the procedure for the practical activity.

9. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack within the specified time frame set by AB.

10. Minimum duration of Assessment of each QP shall be 8hrs/batch (max. 30 candidates).

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ICE/CON/N0601.Implement Tunnel Site Safety Management System	30	60	-	10	100	15
ICE/CON/N0602.Conduct Hazard Identification, Risk Assessment & Control for Tunnelling	30	60	-	10	100	15
ICE/CON/N0603.Manage Ventilation, Gas Monitoring and Occupational Hygiene in Tunnels	30	60	-	10	100	15
ICE/CON/N0604.Emergency Preparedness, Rescue & Evacuation in Tunnel Works	30	60	-	10	100	15
ICE/CON/N0605.Manage Ground Support, Excavation Face Safety and Rockfall Controls	30	60	-	10	100	15
ICE/CON/N0606.Manage Plant, TBM, Mechanical & Blasting Safety in Tunnelling	30	60	-	10	100	15



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ICE/CON/N9901.Implement Safe Work Practices and Environmental Stewardship at Construction Sites	30	60	-	10	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	230	450	-	70	750	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.